

Assignment 1 (15%): Due Friday Week 4, 11:55pm

March 24, 2025

This assessment will cover content taught in the week 1-3 Workshops and Applied classes.

Each task will be graded on;

- Correctness of code & math
- Quality of results and presentation
- Quality of analysis and reasoning

A separate rubric will be provided for a more detailed breakdown of the grading requirements. Submit a .pdf with all the results produced and your analysis of said results as well as runnable Python, MATLAB or Julia code, which should reproduce the results you submit in the .pdf file.

Part 0: Problem Selection (1 Mark)

In this assignment we will develop some of the skills that a computational scientist will use.

Imagine that a client of yours has asked you to develop a method for numerically estimating a real world phenomena. Select one real world phenomenon described by a closed-form function with exactly one independent variable. The function *must* include *at least one* of the following elementary functions; sin, cos, tan, exp, log. Some potential examples include, but are not limited to;

- The damped pendulum function: $\theta(t) = \theta_0 e^{-\gamma t} \cos(\sqrt{\omega^2 - \gamma^2} t)$
- The continuous logistic growth function: $P(t) = \frac{K}{(\frac{K}{P_0} - 1)e^{-rt} + 1}$
- Tide hight: $h(t) = h_0 + A_1 \cos(\omega_1 t + \phi_1) + A_2 \cos(\omega_2 t + \phi_2)$
- Newton's law of cooling: $T(t) = T_{env} + (T_0 - T_{env})e^{-kt}$

Tell us about the function that you chose. Why is it interesting to you?

(Note: Any of the examples given above is a valid choice. You do not need to choose phenomena from outside this list in order to get full marks.)

Part 1: Exact Solution (2 Marks)

Select a region of interest from the domain and produce **two plots** showing how the behaviour of the chosen system changes as the exogenous variables change. What insights can you draw from these plots about the behaviour of the system? Justify and explain your choices and results.

Part 2: Introducing Modelling Errors (3 Marks)

The computer in which your code will be deployed is limited. It cannot actually compute the sin, cos, tan, exp, log functions and must instead approximate them with a Taylor Series Expansion, truncated after the first two nonzero terms, centred on some point in the region of interest you chose in Part 1.

You wish to know how much your model's accuracy will be impacted before it is deployed, and so have decided to simulate the result on your computer. Modify your code from part 1 to take this into account. Explore the same space you did in part 1 with new plots. Explain how and why does the behaviour change?

Part 3: Introducing Data Errors (3 Marks)

The situation gets worse! The computer your code will be deployed on is also limited in its precision. All floating point numbers will be rounded by chopping to 3 digits of precision (in base 10).

Once again, modify your code from part 1 (**not** part 2) to take this into account. Explore the same space you did in part 1 with new plots. How and why does the behaviour change?

(Hint: after each computation, make sure to round the result)

Part 4: Total Error (3 Marks)

You now wish to know how bad things will be when both the data and computational errors are taken into account. Combine your approximations for part 2 and part 3 and investigate the result. Explore the same space you did in part 1 with new plots. How and why does the behaviour change?

Part 5: Sensitivity and Conditioning (3 Marks)

You wish to advise users on where your model can be trusted, and where the results should be suspect. To do this you must in some way estimate the conditioning of this problem. How and why does the conditioning of the problem vary over the domain? When should one trust the output of your code?

Will you use the approximate form of the condition number;

$$CN(f(x)) \approx \left| \frac{xf'(x)}{f(x)} \right|$$

Or the exact form;

$$CN(f(x)) = \left| \frac{\frac{f(x) - \hat{f}(\hat{x})}{f(x)}}{\frac{x - f^{-1}(\hat{f}(\hat{x}))}{x}} \right|$$

and compare the results. If either formulation of the condition number is not appropriate for your function, do not use it, and explain your reasoning for doing so.

AI Statement

Very briefly explain the ways that you have used AI in the production of this assessment.

Explain which AI tools you have used and for what purposes. If you have found and used tools on your own, explain why these tools were selected and provide a URL link to the tool. Note the number of iterations undertaken with each main AI collaborative tool. Describe what output from the tool/service has been included, and where. Summarise how you have altered, adopted, or built on the AI output.

(See Monash Guidelines available at <https://www.monash.edu/student-academic-success/build-digital-capabilities/create-online/acknowledging-the-use-of-generative-artificial-intelligence>)